Chapter 7

Interpretation of Laboratory Data and General Physical Examination by Pharmacists

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Learning Objectives:

Objective 7.1 Describe common laboratory tests and interpret their clinical significance to the practice of pharmacists.
Objective 7.2 Discuss the scope of general physical examination by pharmacists.
Objective 7.3 Explain how to measure and interpret vital signs.
Objective 7.4 Perform a detailed head-to-toe physical examination and explain possible findings.

OBJECTIVE 7.1. DESCRIBE COMMON LABORATORY TESTS AND INTERPRET THEIR CLINICAL SIGNIFICANCE TO THE PRACTICE OF PHARMACISTS

Laboratory tests can provide useful information for clinicians for the diagnosis of a medical condition and for the monitoring of drug therapy (e.g., effect of an antibiotic therapy for a bacterial infection[1]). As a pharmacist, laboratory values can help select the most safe and appropriate therapy for patients, in addition to aid in the monitoring of the selected therapy. For instance, serum creatinine is a laboratory value used to estimate the patient's renal function by calculating creatinine clearance. This information is useful for adjusting the dose of certain medications cleared through the kidneys, such as digoxin. In patients with renal insufficiency, decreased renal excretion of digoxin can lead to increased risk of digoxin toxicity, and as a result, patients may benefit from a dose reduction of digoxin based on creatinine clearance.[2]

The following caution is required while interpreting laboratory values:

- Although laboratory errors are fairly uncommon, they do occur. Potential causes could include technical errors, sample contamination, timing in which the lab value was taken, i.e. trough levels should be drawn just prior to the next dose, and medication interference, i.e. Dopamine and dobutamine influence methods which use the Trinder reaction, thus affecting determination of glucose, cholesterol, and other tests.[41] If any laboratory error is suspected, the test should be repeated.
- Values shown in mmol/L units are equivalent to mEq/L units for some ions when valence is 1, like Hydrogen and Potassium. Otherwise mEq/L could be converted by multiplying mmol/L with valence.
- Always treat the patient, never the laboratory value!

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The following are common laboratory tests:

Electrolytes

Electrolytes and blood chemistries are important for maintaining acid-base and fluid balance. They also play a vital role in nerve and muscle functioning. These usually comprise of sodium, potassium, chloride, calcium, magnesium, and phosphate.[1–4]

Sodium

Sodium is the most prevalent extracellular cation in the body. Its primary function is to regulate the serum osmolality, fluid balance, and acid–base balance. Measuring serum sodium values helps in assessing the patient's electrolyte, water, and acid–base balance. It also helps assess their renal function.[1,5]

Normal values: 135–145 mEq/L.

Clinical Significance An increased level of sodium (hypernatremia) often occurs as a result of dehydration or fluid loss, which could be due to conditions such as gastroenteritis, diarrhea, or Cushing's syndrome.[6] Conversely, hyponatremia is often due to edema from a relative increase in free body water. Certain drugs, like tricyclic antidepressants and loop diuretics, could potentially cause hyponatremia.[6,7]

Potassium

Potassium is the main intracellular cation and plays a key role in many bodily functions including nerve excitability, acid–base balance, and muscle function.[1,8]

Normal values: 3.5–5 mEq/L.

Clinical Significance An increase in potassium (hyperkalemia) could be due to metabolic or respiratory acidosis, or renal failure. Certain drugs, like angiotensin converting enzyme inhibitors (ACEIs) and potassium sparing diuretics, may elevate potassium. Potassium is typically monitored at baseline, and after a few weeks of starting therapy. Meanwhile, the reasons for hypokalemia could include severe diarrhea, respiratory alkalosis, and use of drugs, such as loop and thiazide diuretics as well as osmotic diuretics, like mannitol.[9]

Chloride

Chloride is the principal extracellular anion which functions to serve a passive role in the maintenance of fluid balance and acid–base balance, by having an inverse relationship with bicarbonate.[8,10]

Normal values: 96–106 mEq/L.

Clinical Significance Any deviations in normal values are a sign of fluid or acid–base balance disorder, such as metabolic acidosis, respiratory alkalosis, or prolonged vomiting.[10]

Blood Gas

Carbon dioxide

The majority of carbon dioxide in the plasma is present as bicarbonate ions. However, a small percentage is present in the form of dissolved carbon dioxide. Both carbon dioxide and bicarbonate are extremely important in regulating physiological pH.[11]

Normal values: Bicarbonate ions: 22–28 mEq/L.

Partial Pressure of CO_2 (P_aCO_2): 38–42 mmHg (5.1–5.6 kPa).

Clinical Significance The increase in serum CO_2 levels is seen in metabolic alkalosis, whereas decreasing levels are seen in metabolic acidosis (due to deep breaths to decrease acidosis). Some common causes include the use of diuretic therapy, lactic acidosis, and renal failure.

Blood Glucose

Glucose

Glucose is an important source of energy for most cellular functions, and its regulation is achieved through a complex mechanism comprising insulin, glucagon, cortisol, adrenaline, and other hormones.[12]

Normal values: Fasting blood glucose: 3.9−5.5 mmol/L (70−99 mg/dL).

Clinical Significance High levels of glucose may be detected in patients with Type I, Type II, or gestational diabetes, and are commonly used as a diagnosis of diabetes or pre-diabetes. It is also used as a monitoring tool to monitor the diabetic control of a patient, both for routine monitoring and self-monitoring.[12,13]

Hematology

Full Blood Count

The full blood count provides values for hemoglobin (Hb) and hematocrit (Hct) (to detect anemia), red blood cell (RBC) count, white blood cell (WBC) count, WBC differential count, and RBC morphology.[3,5,14] There are some subtle differences in values between males and females for these measurements, and these need to be taken into consideration when interpreting the values.

Hemoglobin

Hemoglobin is the oxygen-carrying compound in the RBCs and is a direct indicator of the oxygen-carrying capacity of blood.[2,5,15]

Normal values:

Male: 14−18 g/dL or 8.7−11.2 mEq/L.
Female: 12−16 g/dL or 7.4−9.9 mEq/L.

Clinical Significance Increased Hb levels can be due to diseases such as chronic pulmonary lung disease or polycythemia vera. It may also be increased in chronic smokers, those who live at high altitudes (due to low oxygen in air at higher altitudes), and those who engage in regular exercise. The high carbon monoxide content in cigarette smoke prevents oxygen from binding to Hb, making the body interpret low Hb levels, and signal for increased RBCs production. Conversely, lower levels are an indication of anemia or hemorrhage.[15]

Hematocrit

Hematocrit, also known as packed cell volume, describes the volume of blood occupied by RBCs. It can also be used as an indicator of Hb, as hematocrit values are usually three times the value of Hb.[8,11,16]

Normal values:

Male: 39%−49%
Female: 33%−43%

Clinical Significance Similar to Hb, decreased values are an indication of anemia or hemorrhage. The hematocrit can indicate if there is a problem with RBCs, but it cannot determine the underlying cause.

Red Blood Cell Count

RBCs serve to transport oxygen from the lungs to the body tissues. They have a life span of approximately 120 days, before being cleared by the reticuloendothelial system.[17] In most laboratory reports, the results will examine the number of cells per cubic mm, size, shape, color, maturation, and content. These can then be used to classify different types of anemia to pinpoint the etiology. In general, they can be classified by the following:

- RBC size or mean corpuscular volume (indicating average RBC size).
- RBC color or mean corpuscular Hb concentration, which can be hypochromic, hyperchromic, and normochromic.

For example, microcytic, hypochromic anemia could be due to iron deficiency.
Normal values:

Male: 4.3−5.9 × 106 cells/mm^3 or 4.3−5.9 × 10^{12} cells/L.
Female: 3.5−5.0 × 106 cells/mm^3 or 3.5−5.0 × 10^{12} cells/L.

Clinical Significance An increased level of RBCs is usually associated with polycythemia vera, stress, chronic smoking, and living at high altitudes.[8,17] Low RBC counts can be due to anemia, hemorrhage, or chronic renal failure. Low levels of iron causes iron deficiency anemia, folic acid deficiency causes megaloblastic anemia, B-12 vitamin deficiency causes pernicious anemia, and pyridoxine or copper deficiency causes sideroblastic anemia. Long term use of the oral hypoglycemic agent, metformin, has been linked with vitamin B12 deficiency.

White Blood Cell Count

WBCs, or leukocytes, consist of five main types: neutrophils, lymphocytes, monocytes, eosinophils, and basophils. The WBC count is usually given with a breakdown of the percentage of each WBC type. WBCs protect the body against foreign bodies and infection. Neutrophils are the most abundant WBCs.

Normal values: 3.2−11.3 × 10^9 cells/L.

Clinical Significance An increase in WBC count (leukocytosis) is suggestive of an invading organism. In general, the most commonly observed readings are the lymphocyte counts, as these are usually an indicator of viral infections. Meanwhile, an increase in eosinophil levels is taken to be associated with allergic reactions and parasitic infections.[18] Neutrophil levels may rise due to drugs like steroids, infection, or intense exercise.

Platelets

Platelets are a critical element of blood clot formation. They bind together when damaged blood vessels are recognized.[19,20]

Normal values: 150,000−450,000/μL.

Clinical Significance High levels of platelets (thrombocytosis) can be caused by infections and chronic inflammatory disorders, as well as iron deficiency anemia. Decreased platelet counts (thrombocytopenia) may occur in patients with an autoimmune disorder, or while under chemotherapy, due to bone marrow suppression. Heparin can also cause an idiosyncratic reaction of low platelet counts called heparin-induced thrombocytopenia.[21]

Prothrombin Time

A fundamental understanding of the coagulation pathway is imperative in interpreting prothrombin time (PT). The PT is a measure of the integrity of the extrinsic and final common pathways of the coagulation cascade. It consists of tissue factor and factors II (i.e., prothrombin), V, VII, X, and fibrinogen. The test is performed by the addition of calcium and thromboplast, an activator of the extrinsic pathway, to the blood sample; then, the time (in seconds) is measured for the formation of fibrin clot.[22]

Normal values: 9.5−13.5 seconds.

International Normalized Ratio

The international normalized ratio (INR) is a standardized method of reporting the effects of oral anticoagulants, such as warfarin, on blood clotting. This test is commonly performed in individuals receiving warfarin, to ensure the dose is sufficient to prevent thrombosis, and to minimize the risk of bleeding.

Normal values: The reference range for INR should be less than 1.3.[23] An INR range of 2.0−3.0 is the target therapeutic range for most patients on anticoagulant therapy.[24]

Clinical Significance There are various conditions that may affect PT and INR. Some of the causes of a prolonged PT include anticoagulant therapy (e.g., warfarin). Vitamin K deficiency can elevate the INR, and may occur due to malnutrition, biliary obstruction, malabsorption, use of certain broad-spectrum antibiotics, liver diseases (i.e., synthesis of clotting factors is diminished), fibrinogen abnormalities (e.g., hypofibrinogenemia, afibrinogenemia, dysfibrinogenemia), or dilution of plasma clotting proteins after a blood transfusion. Examples of antibiotics that may interact with warfarin, and alter the INR, include cephalosporins, metronidazole, macrolides, fluoroquinolones, penicillins, sulfonamides, and tetracyclines. Furthermore, azole antifungals are most likely to increase bleeding risk in older continuous users of warfarin.[42] It is worth repeating the test to rule out an error if a prolonged PT and INR is observed in patients who are not on anticoagulant therapy or no history of medical problems can explain the abnormal finding. A decreased PT can be caused by vitamin K supplementation, high intake of food containing vitamin K (e.g., liver, broccoli, kale), estrogen-containing drugs (e.g., contraceptive pills, hormone replacement therapy), and freshly frozen plasma transfusion. Patients on warfarin therapy should be counseled regarding major changes in vitamin K rich food, like green leafy vegetables, and its affect on PT/INR.

Partial Thromboplastin Time and Activated Partial Thromboplastin Time

A test similar to the PT, the partial thromboplastin time (PTT) and activated partial thromboplastin time (aPTT), is performed to detect clotting abnormalities. The aPTT is similar to PTT, but an activator is added to speed up the clotting time, thus rendering it more sensitive than PTT due to the narrow range.

Normal values: 60−70 seconds for PTT; 30−45 seconds for aPTT (depending on methods).

Clinical Significance A prolonged PTT or aPTT may indicate congenital deficiencies of intrinsic system clotting factors such as factors VIII, IX, XI, and XII including hemophilia A and hemophilia B (Christmas disease), an inherited bleeding disorder, Von Willebrand disease (which affects platelet function owing to decreased Von Willebrand factor activity). Liver cirrhosis, vitamin K deficiency, and heparin therapy may also prolong PTT or aPTT (in a new sentence). Early stages of disseminated intravascular coagulation, extensive cancer, such as ovarian, pancreatic, or colon cancer, or an acute-phase response leading to high factor VIII levels, may lead to a decreased PTT or aPTT.

There are also some factors that may interfere with the aPTT test. These could be a drug, such as antihistamines, chlorpromazine, heparin, and salicylates, or other factors such as erroneous blood-to-citrate ratio and blood samples drawn from the heparin lock or a heparinized catheter. It is worth considering the confirmation of the test value with subsequent samples to rule out any abnormal results.

Renal Function

The kidneys are responsible for maintaining the homeostasis within the body, through the excretion of solutes and water. The kidney function is affected by the cardiovascular, endocrine, pulmonary, and central nervous systems. Therefore, any abnormalities in these systems may be reflected in renal or urine tests.

Serum Creatinine and Creatinine Clearance

Creatinine is a metabolic product of the dephosphorylation of creatine phosphate from the muscle. It occurs at a relatively constant rate (assuming stable muscle mass) and is excreted primarily by the kidneys. As such, any increase in serum creatinine can be used as a marker to identify patients with renal dysfunction, as it indicates a diminished ability of the kidneys to filter creatinine. In most laboratory reports, the creatinine clearance is also provided, as it is a better indication of kidney function compared to serum creatinine alone. However, a full, true measurement of creatinine excreted by the kidneys requires the collection of urine for 24 h. Alternatively, the creatinine clearance is commonly estimated from the serum creatinine using the Cockroft−Gault formula. Glomerular filtration rate (GFR) calculations also use serum creatinine values. Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation and Modification of Diet in Renal Disease (MDRD) are common formulas to calculate GFR.

Normal values of serum creatinine: 53−106 μmol/L or 0.5−1.5 mg/dL.

Clinical Significance As mentioned previously, high levels of serum creatinine can be used 'to indicate' (or) 'as a marker' would sound better that the kidneys are not functioning optimally. Underlying causes for this could include kidney damage, dehydration, or an obstructed urinary tract. In addition, decreased kidney function can be a microvascular complication of diabetes or can be due to the use of nephrotoxic drugs, such as nonsteroidal antiinflammatory drugs, vancomycin, or contrast media.[25]

Blood Urea Nitrogen

Blood Urea Nitrogen (BUN) measures the amount of urea nitrogen in the blood. Urea nitrogen is excreted from the blood by the kidneys, thus high BUN levels could indicate kidney damage. However, it is a less sensitive marker of renal failure, as it can be affected by hydration, as well as dietary protein intake and protein catabolic rate.

Normal values: 2.1−7.1 mEq/L.

Clinical Significance BUN levels may be high due to heart failure, dehydration, or a high protein diet. Low BUN levels could be due to liver disease. However, it is normal for a pregnant woman to have low BUN levels in the second or third trimester. Blood urea nitrogen to creatinine ratio is used as an indicator of renal function.

Urinalysis

Urinalysis is a useful laboratory test that enables a clinician to identify patients with a wide variety of disorders, especially those associated with the renal system. It is an inexpensive and noninvasive procedure. In general, the test provides a report on the gross examination (e.g., color, clarity), pH, specific gravity, and the presence of protein, glucose, ketones, blood, bilirubin, leukocyte, or nitrates in the urine. Normal urine color should range from clear to dark yellow, with some cloudiness due to the presence of phosphates or urates. If urine is reddish orange, this may indicate the presence of blood or be due to drugs such as rifampicin. Other drugs that may cause urine discoloration include, amitriptylline, which can cause a blue-green color, and chloroquine and metronidazole which can cause brown discoloration of urine. [6]

The kidney normally prevents larger protein molecules from escaping into the urine. Small traces of protein are a common clinical finding, and have no implications. However, the presence of protein (proteinuria) greater than 150 mg/day indicates dysfunction or renal disease. Normally, glucose is actively reabsorbed into the body by the kidney up to a maximum threshold (approximately 180 mg/dL blood glucose). As such, a high level of glucose in the urine is an indication of diabetes mellitus or the need for improved glucose control.[12]

Liver Function

The liver is the largest solid organ in the human body. It plays a central role in all of the body's biochemistry. Thus, liver function tests can give a clearer picture of any of the functions of the liver: synthesis, excretion, or detoxification. These tests can also help indicate liver injury.[1] Elevations in serum transaminases are also seen in patients with myocardial infarction.

Serum Bilirubin

Bilirubin is a reddish-yellow pigment which is produced during the normal breakdown of RBC. It is excreted bile, into the duodenum. Bilirubin circulates in the bloodstream in two forms:

1. Indirect (unconjugated) bilirubin, which does not dissolve in water, and thus cannot be measured. As such, it is derived from both the total and direct bilirubin measurements. It travels through the blood to the liver, where it is changed into a soluble form.
2. Direct (or conjugated) bilirubin, which is the conjugated form of bilirubin. The levels can be measured directly from the blood.

Normal values:

Total bilirubin: 1.7−17.1 μmol/L.
Indirect bilirubin: 3.4−12 μmol/L.
Direct bilirubin: 0−3.4 μmol/L.

Clinical Significance Because of the properties of bilirubin, the levels of serum bilirubin can help clinicians identify liver cell damage in a patient.[26] For example, if there was an increase in total bilirubin and indirect bilirubin, but not direct bilirubin, this could be an indication of RBC hemolysis, because the liver produces direct bilirubin, which is unaffected.

Meanwhile, elevated levels of direct bilirubin may be associated with hepatocellular diseases. In cases where there is excessive total bilirubin, patients will exhibit signs of jaundice, which could result in kernicterus in infants and children.

Urine Bilirubin and Bilirubinogen

As discussed previously, conjugated bilirubin is excreted with the bile into the duodenum, where the conjugated bilirubin is converted by bacteria into urobilinogen. Most of the urobilinogen is excreted in the feces, but some is reabsorbed into the blood and goes either into the liver for reexcretion in the bile, or the urine. In patients with complete bile duct obstruction, no urobilinogen is formed. Therefore, the stools will become gray−white or clay-colored due to the lack of bilirubin. In addition, as conjugated bilirubin cannot be excreted into bile, it will be reabsorbed into the bloodstream and thus spilled into the urine, resulting in high levels of direct bilirubin in the blood.[1] Nevertheless, false-positive results may occur in patients is taking phenazopyridine or phenothiazines. Thus, it is important for pharmacists to note this, since both the drugs, and bile pigments in urine can cause an orange discoloration.

Alkaline Phosphatase

Alkaline phosphatase (ALP) is an enzyme produced mainly in the liver and bone, but can also be found in the kidneys, intestines, and placenta. It is excreted by the liver into the bile, and thus, it is a sensitive indicator of biliary obstruction.

Normal values: 30−120 U/L (depending on age and assay).

Clinical Significance Increased concentrations of alkaline phosphatase may indicate hepatitis, liver cirrhosis, hepatic cancer, or gallstones and high ALP levels may indicate an overreactive parathyroid gland, rickets, Paget's disease, or bone cancer.

Aminotransferases: Aspartate Aminotransferase and Alanine Aminotransferase

These enzymes are located inside the hepatocytes, and function primarily to assist with various metabolic pathways. Any injury to these tissues will release the aminotransferases into the systemic circulation and result in their elevation. Of the two enzymes, alanine aminotransferase is considered a more specific marker of liver disease than aspartate aminotransferase. As such, they are commonly used to assess for hepatocellular injuries, such as hepatitis.

Normal values: 0−35 U/L (depending on age and assay).

Clinical Significance Increased concentrations are associated with hepatitis, and alcoholic liver diseases. Elevated levels could also be seen with drug toxicity or interference with drugs such as acetaminophen, erythromycin, or levodopa.

Thyroid Function

The thyroid is a small, butterfly-shaped gland found in the lower-front part of the neck. This hormonal gland plays a role in regulating many of the body's processes such as metabolism, growth, and maturation of the body.[1] The production of hormones is regulated based on the need of the body. The gland produces three hormones, namely:

1. Triiodothyronine or T3
2. Tetraiodothyronine also called thyroxine or T4
3. Calcitonin

The two main hormones are T3 and T4, produced in the follicular epithelial cells of the thyroid, with iodine being one of the major components of both hormones.

Thyroid-Stimulating Hormone

The thyroid-stimulating hormone (TSH) test, and the T4 test, are the most common thyroid function tests. The TSH test measures the level of circulating TSH, and is used to screen for, and diagnose thyroid disorders. It is also used for routine monitoring during hypothyroidism treatment.

Normal values: 0.3−5 mIU/L.

Clinical Significance If the thyroid gland fails to produce a sufficient amount of these hormones, patients may develop a condition known as hypothyroidism; subsequently, they may experience symptoms such as weight gain, lack of energy, and depression. On the contrary, if the thyroid gland produces too much of these hormones, (i.e., hyperthyroidism) patients may experience weight loss, high levels of anxiety, and tremors. When the thyroid gland is enlarged, it becomes visible as a "goiter," and patients may experience a sense of pressure in the neck, and difficulty in swallowing. If the thyroid becomes even bigger, the tissue enlarges downward, due to restrictions of space. When this occurs, it may lead to more serious effects, such as constriction of the windpipe, causing difficulty in breathing.

Triiodothyronine

T3 is the more potent form of circulating thyroid hormone. It is usually used in the diagnosis of hyperthyroidism.

Normal values: 1.2−3.1 mIU/L.

Clinical Significance High levels of T3 are usually seen in patients with hyperthyroidism, as well as in those with Graves' disease. Conversely, decreased T3 levels are associated with hypothyroidism and/or malnutrition. Some drugs, such as corticosteroids and propranolol, which decrease the conversion of T4 to T3, may result in reduced levels of T3. It is important to take into consideration the effect of amiodarone on thyroid function. Amiodarone, due to its high iodine content, may induce hypothyroidism or thyrotoxicosis. Thus amiodarone should be avoided in patients on medications to manage thyroid abnormalities.[43]

Lipid Profile

The lipid profile is commonly used as part of a cardiac risk assessment, to determine an individual's likelihood of heart disease, and to assist in decision-making on what treatment may be best. When a lipid profile is called for, it primarily comprises the measurement of total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), and triglycerides (TGs). It is recommended that healthy adults with no other risk factors for heart disease be tested with a fasting lipid profile once every 4 to 6 years.[27] If there are other risk factors, or if a high cholesterol level was detected in the previous testing, more frequent testing with a full lipid profile is recommended. Some of the risk factors other than high LDL-C are cigarette smoking, sedentary lifestyle, being overweight or obese, hypertension, diabetes or prediabetes, and preexisting heart disease.[27,28]

As the level of cholesterol and TGs are commonly reported either in mmol/L or mg/dL, the following conversion factors should be used:

mg/dL cholesterol = mmol/L × 38.6.
mg/dL triglycerides = mmol/L × 88.5.

Low-Density Lipoprotein Cholesterol

Approximately 60%−70% of cholesterol in the body is carried as LDL-C in the blood. Lipoproteins are essential for the transportation of cholesterol, which in turn is vital for the biosynthesis of bile acids, vitamin D, and steroid hormones. Dietary intake and endogenous hepatic production are the two primary sources of cholesterol. Metabolism of ingested cholesterol produces very low-density lipoprotein (VLDL) and intermediate-density lipoprotein. Subsequent metabolism of the VLDL produces LDL, which is one of the major components in the development of an atherosclerotic plaque. Most clinicians monitor LDL levels to prevent the progression of coronary artery disease (CAD). The Friedewald formula, used to calculate the LDL level in the blood, is as follows:

LDL = Total cholesterol − HDL − (Triglycerides/5)

However, it is important to bear in mind that the Friedewald formula for estimating LDL is invalid in the following three conditions:

1. presence of chylomicrons,
2. TGs greater than 400 mg/dL, and
3. presence of dysbetalipoproteinemia (type III hyperlipidemia).

These three conditions lead to an underestimated LDL, and in such cases, a direct level is warranted.[29]

Normal values: Values for LDL-C can be divided into several categories:
Optimal: <100 mg/dL or < 2.6 mmol/L.
Near optimal/above optimal: 100−129 mg/dL or 2.6−3.3 mmol/L.
Borderline high: 130−159 mg/dL or 3.4−4.1 mmol/L.
High: 160−189 mg/dL or 3.1−4.9 mmol/L.
Very high: >190 mg/dL or > 4.9 mmol/L.

Clinical Significance The target LDL for the prevention of atherosclerotic plaque formation is between 50 and 70 mg/dL (1.3−1.8 mmol/L).[27,28] A higher value is associated with an increased risk for the development of CAD, and intervention may be needed. It is based on the Framingham Heart Study, which was the first study to reveal a positive association between TC and CAD. A target LDL value of less than 100 mg/dL is particularly imperative in patients with other risk factors, such as cigarette smoking, hypertension, low HDL, and a family history of CAD, as these will accelerate the development of CAD.

High-Density Lipoprotein Cholesterol

HDL-C is produced and secreted by the liver and intestine. It consists mostly of cholesterol, phospholipid, and protein. HDL-C acts as transporters for cholesterol from tissues to the liver. Owing to the reverse cholesterol transport process, it confers a "clean-up" function. This process is called reverse cholesterol transport, because cholesterol synthesized in peripheral tissues is ultimately returned to the liver to be disposed from the body. High levels of HDL-C are deemed desirable due to their inverse relation with coronary risk. Hence, HDL-C is commonly called "good cholesterol". Because of this inverse relation with the incidence of atherosclerosis.

Normal values:

Low: <40 mg/dL or < 1.3 mmol/L.
Optimal: >60 mg/dL or > 1.5 mmol/L.

Clinical Significance In general, elevated HDL levels are ideal, because they are associated with a decreased risk for cardiovascular diseases. HDL levels can be decreased in association with recent illness, starvation, stress, smoking, obesity, lack of exercise, hypertriglyceridemia, or may be due to some medications (e.g., thiazide diuretics, steroids, beta blockers), and regular aerobic exercise, smoking cessation, a decrease in body mass index, and mild therapy with HMG-CoA reductase inhibitors (i.e., statins), may increase HDL-C levels. The slight rise in HDL-C levels from these drugs may be attributable to inhibition of Rho-signaling pathways, with activation of peroxisome proliferator-activated receptor alpha. Increases in HDL-C levels may also be seen due to decreasing plasma cholesteryl ester transfer protein activity by statins.[27,28]

Triglycerides

TGs are lipid compounds made up of a glycerol, which is esterified to three fatty acid chains of varying length and composition. It is the most abundant dietary lipid compound found in the daily diet.

Normal values: There are currently no targets for serum TG levels, but levels below 150 mg/dL <1.7 mmol/L are considered desirable.

Clinical Significance Some substances and drugs that may account for TG elevation are ethanol, corticosteroids, non-cardioselective beta blockers, thiazide diuretics, bile-binding resins, oral estrogens, progestins, tamoxifen, and antiretroviral therapy. Patients who are at risk for developing secondary hypertriglyceridemia in the setting of CAD, diabetes,

disease states secondarily elevating TGs, or another CAD? should receive routine screening. Patients who have consumed a meal high in lipids, and have not fasted for 8 hours before a cholesterol test, may show false-positive highTG levels. In addition, patients who have recently consumed ethanol may show an elevation of TGs that may not certainly be indicative of baseline levels.

Cardiac Enzymes

Creatinine Kinase

Formerly known as creatinine phosphokinase, creatinine kinase (CK) is a key protein enzyme in catalyzing the reversible phosphorylation reaction. CK levels can be further divided into its isoenzymes, depending on where they originate from; muscle (CK-MM), brain (CK-BB), and cardiac tissue (CK-MB). Making up 15%−30% of the total CK in the heart muscle, CK-MB is found mostly in the myocardium, whereas a much smaller proportion is found in the skeletal muscle. CK-MB is sensitive and very specific to myocardial injury.

Normal values: The normal range of CK-MB is 3%−5% of total CK.

Clinical Significance Following the onset of acute myocardial injury, CK-MB begins to rise in 4−6 h, and returns to baseline level after 36−48 h. Therefore, CK-MB is also a good marker of acute myocardial injury, reinfarction, or infarct extension. CK-MB can be used to indicate successful reperfusion after fibrinolysis, estimate infarct size, and predict infarct-related mortality.

Troponin

Troponins are protein molecules, and are part of the cardiac and skeletal muscle. Without cardiac myocyte damage, cardiac troponin levels do not increase in the presence of skeletal damage. It is unlike CK-MB, which can be increased on injury to these tissues, potentially leading to false-positive results. Hence, troponin testing is more reliable than CK-MB testing in that sense.

Clinical Significance Troponin values are usually assessed on clinical suspicion of myocardial infarction (MI). In healthy patients, troponins are undetectable, although this may not be the case as more sensitive assays become available. In a patient with presentations such as chest pain and possible MI, an abnormal value is that above the 99th percentile of the healthy population. In patients with MI, a time lag is seen before troponin elevations can be detected. Therefore, the markers should be serially monitored, should there be suspicion for the acute coronary syndrome. Normally three sets of cardiac enzymes are checked. The initial recommendation is to assess the markers every 3 hours until the expected peak is reached.[30]

It is important to note some of the analytical interfering factors that may lead to falsely elevated troponin results. Examples of such items include a specimen with incomplete clotting (often seen in patients with coagulopathy or those on anticoagulant therapy), elevated bilirubin levels, markedly elevated alkaline phosphatase levels, immunocomplex formation, and analyzer malfunction.

Information obtained from lab tests can help pharmacists make informed decisions by individualizing therapy depending on the patient's current conditions. Therapy can be monitored throughout the course of treatment, and alterations can be made to give the appropriate drug regimen.

OBJECTIVE 7.2. DISCUSS THE SCOPE OF GENERAL PHYSICAL EXAMINATION BY PHARMACISTS

The general physical examination is a systematic process by which the clinician uses his/her sensory skills to evaluate physical signs pertaining to a patient's clinical condition, and then correlates that information with previous patient history for the formulation of a clinical diagnosis.[31] It is essential that the examiner has a sound knowledge of anatomy, physiology, and pathophysiology to identify and evaluate clinical signs and symptoms.

Integrating physical assessment skills into pharmacy practice is important for improving the quality of patient care. Data obtained from a physical assessment can allow the pharmacist to monitor and optimize medication management, to determine whether a medication is appropriate, effective, and safe. Information from a physical assessment will also allow a clinician to screen patients at risk for chronic conditions, such as hypertension, and help set the direction of the

therapeutic plan. Practicing physical assessment skills will also promote better communication among healthcare practitioners and improve our understanding of patient care.

Context

The clinical examination should be part of a process wherein the clinician has the patient's basic history available at his or her disposal. The goal of such an assessment is to gather pertinent information to identify underlying clinical conditions and set the direction of a therapeutic plan. The clinical examination begins well before talking to the patient. A mere look at the patient by observing gait, level of consciousness, skin color and obvious lesions, grooming, facial expression, body odors, behavior, speech, and overall presentation would help identify many important findings.[31,32] A further detailed evaluation would systematically focus on every organ system to identify the clinical processes occurring beneath the surface.

There are four techniques that are typically used during a physical assessment: (1) inspection, in which each body system is assessed for any deviations using vision, smell and hearing, (2) palpation, in which you need to use different parts of your hands in varying degrees of pressure to touch the patient, (3) percussion, in which you sharply and quickly tap your fingers or hand against specific body parts of the patient, and (4) auscultation, in which a stethoscope is used to listen to various organ sounds.[32]

Inspection is also known as "concentrated watching," and it involves observing the patient for physical signs. For instance, yellowing of the skin can be a sign of jaundice. The inspection begins the moment the patient is introduced; good lighting is important for this first step. The patient can be used as his or her control by comparing the right and left sides of the body.

Palpation involves using the sense of touch to collect information. For example, the radial pulse can be determined by palpating the radial artery. The fingertips are best for detecting fine tactile discrimination such as skin texture. The fingers and thumbs can be used to grasp and detect organ position, shape, and consistency. The backs of the hands and fingers are best for determining skin temperature, and the base of the fingers or ulnar surface of the hand can be used for detecting vibrations.

Percussion involves tapping the patient's skin with short and sharp strokes to assess the density of structures 4−5 cm below the skin. These strokes result in vibrations that can produce a characteristic sound to map the location and size of the organ and to determine whether air, fluid, or a mass exists beneath the surface. For instance, the disappearance of a dull sound (indicating fluid accumulation) over the lung can indicate a sign of antibiotic effectiveness.

Finally, auscultation can be performed to assess sounds originating within the organ or body cavity, such as heart sounds, bowel sounds, etc. A stethoscope is required to perform auscultation.

Pharmacist−Patient Interaction

For a clinician, a medical examination is a learned behavior. However, for a patient, it is a first-time experience, where someone is handling his or her body systems. It is highly essential that the clinician generates trust and confidence of the patient to avoid any embarrassment to the patient, and also to help bring out the clinical signs. It is essential that the patient be made as comfortable as possible. The clinician has to be polite and genuinely compassionate when approaching the patient. There should be a continuous conversation during the examination to relieve any anxiety the patient might have during the process. A brief description of the findings to the patient will help him/her understand the physical examination process and to participate in the clinical management plan.

Prerequisites

A trained and knowledgeable mind is the most important prerequisite for a good clinical examination. Some basic equipment, if made available, will facilitate detailed assessment. This would include, and is not limited to, a light/torch, measuring tape, blood pressure monitoring device, tongue depressor, cotton, stethoscope, otoscope, fundoscopy, gloves, magnifying glass, thermometer, and a tuning fork.

As the environment affects the quality of the physical examination, it is wise to arrange for a quiet area for privacy, darkening the room for parts of the examination, and providing comfort for the patient and examiner. Although there is no right or wrong sequence, it is important to follow a systematic approach to ensure full completeness of the examination. As a convention, the examiner would stand on the right side as the patient is lying on a couch. In the sitting position, it is appropriate to stand in front of the patient and move around with various steps of examination.

OBJECTIVE 7.3. EXPLAIN HOW TO MEASURE AND INTERPRET VITAL SIGNS

Body Temperature

Body temperature is regulated by the hypothalamus to maintain the core temperature of approximately 37°C (98.6°F).[31,32] Body temperature can be affected by diurnal fluctuations throughout the day (e.g., lowest in the morning, highest in the late afternoon to early evening), hormones, exercise, smoking, consumption of hot or cold beverages, and age.[32] Measurement of body temperature can be obtained by an oral, rectal, axillary, or tympanic thermometer.

An oral thermometer is ideal for alert patients. The body temperature can be obtained by placing the oral thermometer under the tongue in either of the sublingual pockets with the lips closed. Hot or cold liquids and smoking can alter the temperature reading. As such, it is advised to delay the temperature measurement by 10−15 min after consuming beverages or smoking. The rectal thermometer may be considered in comatose patients or patients who are unable to close their mouth (e.g., intubation, wired mandible, facial surgery). The temperature reading through the rectal route may be 0.5°C (0.9°F) higher than the oral reading. The axillary route may be ideal for infants and small children. This reading may be 0.5°C (0.9°F) lower than the oral route, and it is obtained by placing the thermometer under the arm into the center of the axilla with the patient's arm folded over the chest to keep it in place for 5−10 min. A tympanic thermometer can be considered for unconscious patients, emergency departments, and labor and delivery units. These readings may be 0.8°C (1.4°F) higher than the oral route. Temperature readings are obtained by gently pulling the ear up and back to straighten the ear canal (if under 3-years old, the ear is pulled downward and back) and placing the probe into the ear canal. Most digital tympanic thermometers can provide a reading in 2−3 s. Additionally, noncontact forehead thermometers are also available and may be more advantageous for reading temperatures in children.[33,34]

A normal body temperature reading is 35.8−37.3°C (96.4°F to 99.14°F).[31,32] A higher than normal temperature reading can be a result of pyrexia (i.e., fever) caused by infection, tissue breakdown (e.g., MI), or neurological (e.g., brain tumor). A high-temperature reading can also be a result of hyperthermia, in which heat production or external heat exposure exceeds heat loss, and the body is unable to thermoregulate. This can occur due to heat stroke, drugs (e.g., serotonin syndrome), and hyperthyroidism. A lower than normal temperature reading can be a result of hypothermia in which heat loss exceeds heat production. This can occur due to prolonged exposure to cold or intentional induction (e.g., cardiovascular surgery).

Peripheral Pulses

A quick assessment of all peripheral pulses would help assess heart rate, rhythm, and force. Peripheral pulses include radial, brachial, carotid, popliteal, posterior tibial, and dorsalis pedis pulses. The peripheral pulse is felt by the gentle compression of the artery against the underlying structures such as bone or the soft tissues. The radial pulse is felt by palpating the artery at the lower end of the radius bone anteriorly. The brachial artery can be felt at the cubital fossa just medial to the biceps tendon. A posterior tibial pulse is felt 1 inch below and behind the medial malleoli, against the body of calcaneum, in the groove between the medial malleolus and the Achilles tendon. Dorsalis pedis pulses are felt at the ankle in the first interosseous space of the feet, medial to the external hallucis tendon. While examining the peripheral pulse, it is necessary to appreciate the rate, rhythm (regular vs. irregular), and character (normal vs. weak vs. bounding) of the sound, as well as the condition of the vessel wall (e.g., calcified arteries in elderly can feel like fibrous cords). It is also a good practice to feel both radial pulses simultaneously to appreciate any radio-radial delay (e.g., aortic dissection) and the radial and femoral pulses to appreciate radio-femoral delay (e.g., coarctation of the aorta).

To obtain an accurate reading of the radial pulse, the clinician can count the number of beats in 30 s; if the beats are regular, multiply this number by two to obtain the number of beats per minute (bpm). However, if the beats are irregular, it is recommended to count the number of beats for the full minute. The normal heart rate for adults is 60−100 bpm. Children are expected to have higher heart rate ranges (e.g., newborns: 70−170 bpm, 1- to 6-year-olds: 75−160 bpm, 6- to 12-year-olds: 80 to 120 bpm).

Respiration Rate

The respiratory rate should be measured without the patient's knowledge, as the rate can change as soon as the person is aware that it is being done. To measure the respiratory rate, maintain the position of a radial pulse measurement and observe the patient's chest or abdomen for respiration. One respiration is inhalation and exhalation. To obtain the respirations per minute (rpm), count the number of respirations in 30 s; if the rhythm is regular, multiply this number by 2. However, if the rhythm is irregular, the number of respirations should be counted for the full minute. In adults, the normal

respiration rate is 12−20 rpm, whereas the normal respiration rate for children is between 20 and 30 rpm. A lower than the normal respiratory rate is termed bradypnea, and a higher than the normal respiratory rate is termed tachypnea.

Blood Pressure

Blood pressure is measured in millimeters of mercury (mmHg) and comprises two components: (1) systolic blood pressure and (2) diastolic blood pressure. The systolic blood pressure is the highest pressure that is felt in the arteries, and it occurs when the ventricles contract.[18] The diastolic blood pressure is the lowest or resting pressure that occurs when the ventricles relax. A blood pressure reading can be obtained using an oscillometric (electronic) automated monitoring, or an auscultatory (manual) reading with a mercury or aneroid sphygmomanometer. An ideal oscillometric blood pressure device takes six consecutive readings at one- to two-min intervals.[35] The clinician would attend the first reading to ensure the patient is in the proper position and that the monitor is operating properly. The patient is then left alone for the subsequent readings. The device then discards the first reading and averages the next five measurements to obtain the blood pressure reading. If necessary, an ambulatory (out-of-office) blood pressure device can be worn by the patient to record blood pressure readings over a 24-h period at half-hour to one-hour intervals.[36,37] This allows the blood pressure to be monitored for fluctuations throughout the day based on activity and medications. It also helps rule-out white coat hypertension in clinical settings.

An automated home blood pressure machine can be purchased for patients to use at home. Although these machines are typically easy and convenient to use, the accuracy of the measurement depends on the proper positioning of the patient and placement of the cuff. Moreover, readings can be inaccurate if the heart rate is irregular and if there is physical movement (e.g., shivering).

It is advised to use an automated blood pressure device that has been validated and endorsed by a regulating body.[38] The aneroid and mercury sphygmomanometers have traditionally been used in practice to obtain a manual blood pressure reading. Accuracy depends on the proper standardized technique and calibration of the device.

To obtain an accurate blood pressure measurement, it is important that the patient be seated and comfortable. Ideally, the patient should be well rested for at least 5 min before the blood pressure measurement. As such, the pharmacists could conduct a medication history with the patient during this period. The patient should also not have smoked at least 30 minutes before getting their blood pressure monitored. The patient should subsequently be seated with back supported, feet flat on the floor, and arm slightly bent, palm up, and supported at heart level.[31,39] Next, the appropriate cuff size should be selected. The cuff size can be determined by ensuring the bladder width is approximately 40% of the arm circumference and the bladder length is approximately 80% of the arm circumference. Once an appropriate cuff size has been selected, the center of the cuff bladder can be placed over the brachial artery. The cuff should be wrapped smoothly and snuggly around the arm approximately 2.5 cm above the crease of the elbow. The patient is now ready to have a blood pressure measurement.

An automated device will typically take a measurement after being activated and display the blood pressure reading digitally. However, if using a manual cuff, the manometer should be in direct line of eye site to allow for an accurate reading.

With a manual cuff, it is recommended to estimate the systolic blood pressure first, to determine how high to raise the cuff pressure to prevent discomfort to the patient and to prevent error caused by an auscultatory gap. To estimate the systolic pressure, the clinician palpates the radial pulse and then inflates the cuff to a point at which the radial pulse can no longer be felt. The clinician then adds 30 mmHg to this reading to obtain an estimated systolic blood pressure. After the systolic blood pressure is estimated, the clinician can use the stethoscope, placing the bell of the stethoscope lightly over the brachial artery site, but in a position to make a tight seal with the patient's skin. The clinician inflates the cuff again to the estimated systolic blood pressure, and then slowly deflates the cuff at a steady rate of 2−3 mmHg/s. While the cuff is deflating, the clinician is listening for Korotkoff sounds with the stethoscope. The pressure at which the first of two consecutive beats are heard (Korotkoff Phase I), is the systolic blood pressure. The pressure at which the last beat is heard (Korotkoff Phase V), is the diastolic blood pressure.[31,32] It is recommended to continue listening until 20 mmHg below the diastolic blood pressure, and then rapidly and completely deflate the cuff. The clinician should then wait 2 minutes and repeat the reading. It is recommended to take three readings on the same arm, and average only the last two readings.

Common mistakes while taking a blood pressure reading include using an incorrect cuff size (e.g., if it is too small for the patient it can overestimate the blood pressure reading) and the patient's positioning (e.g., if the patient's arm is below heart level, or if the patient is not rested or comfortable, it can overestimate the reading). Other common mistakes while using a manual device for measuring blood pressure include stopping during deflation or reinflating the cuff too soon/deflating too quickly to allow enough time to hear the Korotkoff sounds.[37] A cuff deflation rate of 2 mmHg per beat is necessary for accurate blood pressure measurement.

There are also several factors that can influence blood pressure. Factors that can raise blood pressure include nicotine or caffeine consumption in the last 30 min, certain drugs (e.g., decongestants, corticosteroids, NSAIDs), exercise, anxiety, full bladder, room temperature, patient talking during a reading, or tight clothing around the forearm. Factors that can lower blood pressure include fasting or certain drugs (e.g., depressants). Some patients have an auscultatory gap, which is a silent interval between the systolic and diastolic pressures caused by arterial stiffness and atherosclerotic disease.[1] It can result in an underestimation of the systolic blood pressure or an overestimation of the diastolic blood pressure.

OBJECTIVE 7.4. PERFORM A DETAILED HEAD-TO-TOE PHYSICAL EXAMINATION AND EXPLAIN POSSIBLE FINDINGS

The entire process of general physical examination will involve inspection and palpation. A general head-to-toe survey is useful to find out any obvious abnormalities. It could include any patient concerns such as pain/discomfort, respiratory distress, skin and pigment changes, abnormal posture of extremities, and the general look of the patient. It is followed by a detailed head-to-toe examination.

Head and Neck

The examination will systematically look for signs in the head and neck, and will try to correlate with anatomic and pathophysiologic processes. A general look at the scalp and facial hair will help in assessing the types of alopecia; the common ones are male- and female-type baldness, alopecia areata, scarring, and anagen and telogen effluvium. Excess of facial hairs in females (hirsuitism) is a sign of androgen excess and can suggest the presence of polycystic ovarian syndrome (more likely) or ovarian/adrenal neoplasms (less likely). Malar rash over the bridge of the nose and cheek are suggestive of immunologic disorders such as systemic lupus erythematosus (SLE). Other abnormalities in the face include scars, chloasma, moles, and acne.

Eyes

A quick look at the eyes is a great source of information about a patient's general health. The presence of pale palpebral conjunctiva signifies anemia. It can be substantiated by examining the tongue, nail beds, or palms for evidence of pallor. Any cause of anemia would produce pallor.

Examination of the bulbar conjunctiva in daylight will help identify any yellowish tinge/discoloration which may signify jaundice. Sometimes, people who have had chronic exposure to sunlight and dust tend to have muddy conjunctiva which can be mistaken for jaundice. In mild jaundice, the lemon yellow tinge is usually visualized only in the eyes and in oral mucosa, in the floor of mouth, after asking the patient to elevate their tongue. In marked elevations of bilirubin, a yellowish tinge can be visualized on the skin, as well as the patient's palms and soles. Patients with obstructive jaundice tend to have a deep yellow icterus and scratch marks over the body due to excess itching, which arises as a result of elevations of bile acids.

Other ocular findings which one must be aware of are subconjunctival hemorrhages (usually self-limiting), Kayser−Fleischer rings (i.e., brownish-yellow rings visible around the corneoscleral junction in Wilson's disease), and conjunctivitis. Drooping of the eyelid can signify oculomotor nerve (i.e., cranial nerve III) palsy. Extraocular movements can be assessed to check the integrity of cranial nerves III, IV (i.e., the trochlear nerve which innervates the superior oblique muscle), and VI (i.e., abducens which innervates the lateral rectus muscle). It is important to take proper medication history, since drugs like ethambutol may cause vision problems ranging from blurred vision to red-green blindness. Sildenafil may cause a mild and transient blue tinge to vision.[44]

Face

Check for any facial asymmetry which can be congenital, due to developmental abnormalities, or acquired, due to facial nerve palsy on one side (e.g., reduced wrinkles, inability to close eyes and flattening of nasolabial folds on the affected side, deviation of angle of mouth to the opposite side). Masked facies is associated with Parkinson's disease, in which facial expressions are lost.

Oral Cavity

After checking for pallor and icterus, evaluate general dental hygiene by looking for any caries or infection. Any oral pigmentation may be a cause of concern, as it could suggest a serious condition, such as melanoma or Peutz−Jeghers syndrome.

Common lesions may also be due to smoking or could be benign nevi. The moistness of the tongue and mucous membrane is a sign of hydration status of the patient and is helpful in assessing the need for fluid in a patient with acute illnesses. Presence of white patches in the oral cavity may indicate oral candidiasis, which could be due to antibiotics or inhaled corticosteroids. In the latter case, the patient should be advised to rinse their mouth after using the corticosteroid.

Neck

Examination of the neck should proceed systematically. The neck should be inspected from the front and sides for any obvious swelling/masses. Common swellings include thyroid enlargement (e.g., diffuse in Graves' disease and nodular in nodular goiter). The patient should be asked to swallow saliva or be given a glass of water to drink, and movement of thyroid upward should be observed which confirms the thyroid swelling. Thyroglossal cyst is also a midline cystic swelling in the region or slightly above the thyroid gland. Such swelling will move up on protrusion of the tongue, whereas thyroid swelling will not. Other common neck swellings can include lymph nodal masses, which can be detected by submandibular gland enlargement just below the mandible, and parotid gland enlargement just under the earlobe. As a convention, it is recommended to palpate the neck with the patient sitting on a chair and the examiner standing behind the patient. Fingertips are used to palpate the neck looking for any lymph nodes in the submental and submandibular region, anterior and posterior triangle of the neck, and supraclavicular fossa. Thyroid gland should be palpated from behind, and movement assessed with swallowing.

With two fingers, careful palpation of the carotid arteries can be done by moving lateral to the thyroid cartilage, and in the groove between the trachea and sternocleidomastoid muscle. Care should be taken not to palpate both the arteries simultaneously. Also in elderly patients, undue pressure should be avoided as it can stimulate carotid sinus and lead to bradycardia and syncope, or can dislodge an atherosclerotic plaque, leading to a cerebrovascular accident. With the patient in 45-degree position, the neck should be examined for a jugular venous pulse, which can be elevated in a patient with congestive cardiac failure and right heart dysfunction. Engorged neck veins such as the external jugular vein is also a sign of right heart dysfunction and biventricular failure.

Upper Limbs and Hands

It is usually recommended that the whole upper extremity be observed at once. Any obvious drooping of the shoulder can be picked up on by examining the patient from behind while they are standing. Any swelling, deformities, and congenital abnormalities should be recorded. A patient can have generalized swelling, which indicates circulatory issues, or localized swelling which suggests fractures, infections, tumors, and tendon abnormalities. Observe the skin color of the hands and fingers. Hyperemia is usually a result of bacterial infection. Dry and shiny skin may occur with systemic diseases such as scleroderma; punctate digital ulcers and nail atrophy are seen in Raynaud's disease. Hypo/hyperpigmentation plus hypertrichosis and dry skin may be signs of neuropathy.

Inspection of the fingernails can also provide information about systemic disorders. Flattening (platonychia) and spooning (koilonychia) are suggestive of severe iron deficiency anemia. Clubbing is usually a sign of a lung disorder, but can also be seen in inflammatory bowel diseases, cirrhosis, cyanotic congenital heart disease, and lung cancers. Half-and-half nail syndrome is a condition in which the proximal portion of the nail bed is white, and the distal portion is dark (e.g., usually brown); this may signify advanced renal disease. It is also seen with advanced HIV infection and lymphomas. Terry's nail is a condition where the proximal 80% of the nail bed is white, and the distal 20% are normal/pink; this is usually seen with circulatory insufficiency. Other abnormalities which can be seen include Mees' lines (i.e., transverse white lines suggestive of arsenic poisoning), Beau's lines (i.e., transverse ridge/depressions due to intermittent cessation of nail growth in systemic illness/trauma/zinc deficiency), blue nails (e.g., drugs—zidovudine), fungal infections, nail pitting due to psoriasis, and splinter hemorrhages due to bacterial endocarditis.

Two distinct nodular swellings are found on the fingers of patients with osteoarthritis. Posterolateral swelling of distal interphalangeal (DIP) joints are called Heberden's nodes, and swelling of proximal interphalangeal (PIP) joints is called Bouchard's nodes.

The patient can also have swelling of the small joints of hands. The swelling is usually associated with stiffness in the fingers on flexion, especially during the early morning, and is usually associated with tenderness on palpation. It is suggestive of polyarthritis which can be due to various causes. The pattern of involvement of the small joint of the hands is quite suggestive of the etiology. Bilateral symmetrical involvement of metacarpophalangeal (MCP) and PIP joints are common with rheumatoid arthritis. Duration of morning stiffness can help distinguish the type of arthritis. Morning stiffness typically lasts < 30 minutes in osteoarthritis, compared to rheumatoid arthritis. Conditions preferentially affecting the DIP

joints include osteoarthritis, psoriatic arthritis, SLE, and reactive arthritis. Arthritis of the first carpometacarpal joint is also seen typically in osteoarthritis; nowadays, with the advent of smartphones, these are more commonly seen even in younger individuals.[40] Deformities of the small joint of the fingers can produce characteristic morphology. This is common in advanced rheumatoid arthritis, which can lead to swan-neck deformity (i.e., flexion of MCP and DIP joints and hyperextension of PIP joints) and Boutonniere deformity (i.e., flexion of proximal interphalangeal PIP joints, an extension of the DIP joint of the fingers, and flexion of the MCP joint. A note should be made of any missing digits due to amputations. Other congenital abnormality such as polydactyly or syndactyly should be noted.

Lower Limbs

On examination, the lower limbs should preferably be exposed to the groin region. Privacy needs to be addressed, and patient consent is mandatory. The limbs should be examined for any obvious deformities, swelling, and signs of inflammation such as redness. Gentle pressure over bony prominences, such as the shin or the malleoli for a few seconds will elicit the presence of pitting pedal edema. This is usually seen in both lower limbs in conditions with an overall increase in total body water (e.g., hypoalbuminemia, CHF, cirrhosis, and nephrotic syndrome), and unilaterally in infections and venous thrombosis. Drugs like thiazolidinediones, estrogens, and amlodipine may cause pedal edema. The health of the toenails also has similar significance as upper limb nails. The inter-digital areas should be examined for the presence of whitish/reddish lesions suggestive of infections (e.g., tinea pedis). Special attention should be paid at the feet of diabetic patients; look for sores, blisters or wounds. Watch out for any staining on socks, and check for numbness or loss of sensation.

CONCLUSION

Confidence in clinical skills like interpretation of laboratory data, patient history interview, and physical assessment, is vital for pharmacists. Clinical significance of laboratory tests indicate relevant pathological and drug-induced changes in biochemistry. A trained and knowledgeable pharmacist is able to comfortably engage with a patient to gather basic history, while screening for signs of chronic disease or medication side effects. A pharmacist can provide optimal patient care by using the right techniques to measure vital signs, and performing a detailed head-to-toe physical examination. Knowledge about the possible findings helps the pharmacist identify the root cause, and provide the best drug therapy. All these skills are practiced with the ultimate goal of improving patient care outcomes. Doesn't fit in Repetition.